Q1 2021 (01 Sep Shift 2)

The ranges and heights for two projectiles projected with the same initial velocity at angles 42° and 48° with the horizontal are $\mathrm{R}_1, \mathrm{R}_2$ and $\mathrm{H}_1, \mathrm{H}_2$ respectively. Choose the correct option :

(1) $\mathrm{R}_1 > \mathrm{R}_2$ and $\mathrm{H}_1 = \mathrm{H}_2$

(2) $\mathrm{R}_1 = \mathrm{R}_2$ and $\mathrm{H}_1 < \mathrm{H}_2$

(3) $\mathrm{R}_1 < \mathrm{R}_2$ and $\mathrm{H}_1 < \mathrm{H}_2$

(4) $\mathrm{R}_1 = \mathrm{R}_2$ and $\mathrm{H}_1 = \mathrm{H}_2$

Q2 2021 (31 Aug Shift 1)

A helicopter is flying horizontally with a speed 'v at an altitude 'h' has to drop a food packet for man on the ground. What is the distance o helicopter from the man when the food packet is dropped?

(1) $\sqrt{\frac{2\mathrm{ghv}^2+1}{\mathrm{h}^2}}$

(2) $\sqrt{2\mathrm{ghv}^2+\mathrm{h}^2}$

(3) $\sqrt{\frac{2v^2h}{g}+h^2}$

(4) $\sqrt{\frac{2gh}{\mathrm{v}^2}+\mathrm{h}^2}$

Q3 2021 (27 Aug Shift 2)

A player kicks a football with an initial speed of $25\ \mathrm{ms}^{-1}$ at an angle of 45° from the ground. What are the maximum height and the time taken by the football to reach at the highest point during motion ? (Take $\mathrm{g}=10\ \mathrm{ms}^{-2}$)

(1) $\mathrm{h}_{\max}=10\ \mathrm{m} \quad \mathrm{T}=2.5\ \mathrm{s}$

(2) $\mathrm{h}_{\max}=15.625\ \mathrm{m} \quad \mathrm{T}=3.54\ \mathrm{s}$

(3) $\mathrm{h}_{\max}=15.625\ \mathrm{m} \quad \mathrm{T}=1.77\ \mathrm{s}$

(4) $\mathrm{h}_{\max}=3.54\ \mathrm{m} \quad \mathrm{T}=0.125\ \mathrm{s}$

Q4 2021 (26 Aug Shift 2)

A bomb is dropped by fighter plane flying horizontally. To an observer sitting in the plane, the trajectory of the bomb is a :

(1) hyperbola

(2) parabola in the direction of motion of plane

(3) straight line vertically down the plane

(4) parabola in a direction opposite to the motion of plane

Q5 2021 (26 Aug Shift 2)

A particle of mass m is suspended from a ceiling through a string of length L. The particle moves in a horizontal circle of radius r such that $r=\frac{L}{\sqrt{2}}$. The speed of particle will be :

(1) $\sqrt{\mathrm{rg}}$

(2) $\sqrt{2\mathrm{rg}}$

(3) $2\sqrt{\mathrm{rg}}$

(4) $\sqrt{\frac{\mathrm{rg}}{2}}$

Answer Key

Q1 (2) **Q2** (3) **Q3** (3) **Q4** (3)

Q5 (1)

Q1 (2)

Range $\mathrm{R}=\frac{\mathrm{u}^{2} \sin 2 \theta}{\mathrm{g}}$ and same for θ and $90-\theta$

So same for 42° and 48°

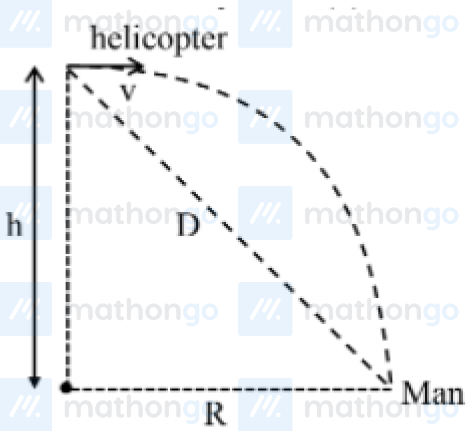
Maximum height $\mathrm{H}=\frac{\mathrm{u}^{2} \sin ^{2} \theta}{2 \mathrm{~g}}$

H is high for higher θ

So H for 48° is higher than H for 42°

Option (2)

Q2 (3)



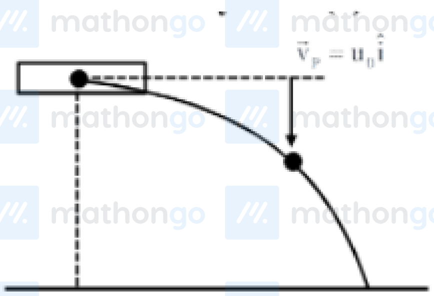
$$
\begin{aligned}
& R=\sqrt{\frac{2 h}{g}} \cdot v \\
& D=\sqrt{R^{2}+h^{2}} \\
& =\sqrt{\left(\sqrt{\frac{2 h}{g}} \cdot v\right)^{2}+h^{2}} \\
& D=\sqrt{\frac{2 h v^{2}}{g}+h^{2}}
\end{aligned}
$$

Option (3) is correct

Q3 (3)

$$
\begin{aligned}
H &= \frac{U^2 \sin^2 \theta}{2\,g} \\
&= \frac{(25)^2 \cdot (\sin 45)^2}{2 \times 10} \\
&= 15.625 \text{ m} \\
T &= \frac{U \sin \theta}{g} \\
&= \frac{25 \times \sin 45^{\circ}}{10} \\
&= 2.5 \times 0.7 \\
&= 1.77 \text{ s}
\end{aligned}
$$

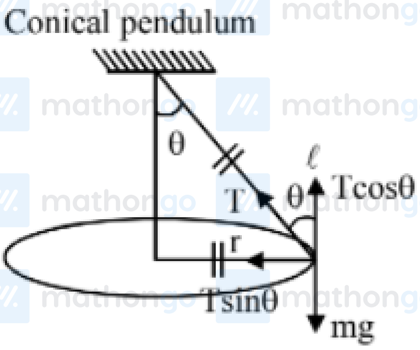
Q4 (3)



$$
\begin{aligned}
v_B &= u_0 \hat{i} - gt\hat{j} \\
\vec{v}_{B/P} &= \vec{v}_B - \vec{v}_p \\
\vec{v}_{R/P} &= -8t\hat{j}
\end{aligned}
$$

straight line vertically down

Q5 (1)



$r = \frac{\ell}{\sqrt{2}}$

$\sin\theta = \frac{r}{\ell} = \frac{1}{\sqrt{2}}$

$\theta = 45^\circ$

$T\sin\theta = \frac{mv^2}{r}$

$T\cos\theta = mg$

$\tan\theta = \frac{v^2}{rg} \Rightarrow v = \sqrt{rg}$

Ans. 1